Lesson 14: Raising the Stakes

001 Game Creator (Windows only)

VIDEO GAME CREATION

This week: later waves get genuinely harder - more enemies, moving faster - using the exact wave variable you've had since Lesson 11.

Open your own [YourName]Game project from Lesson 13 - don't start a new one.

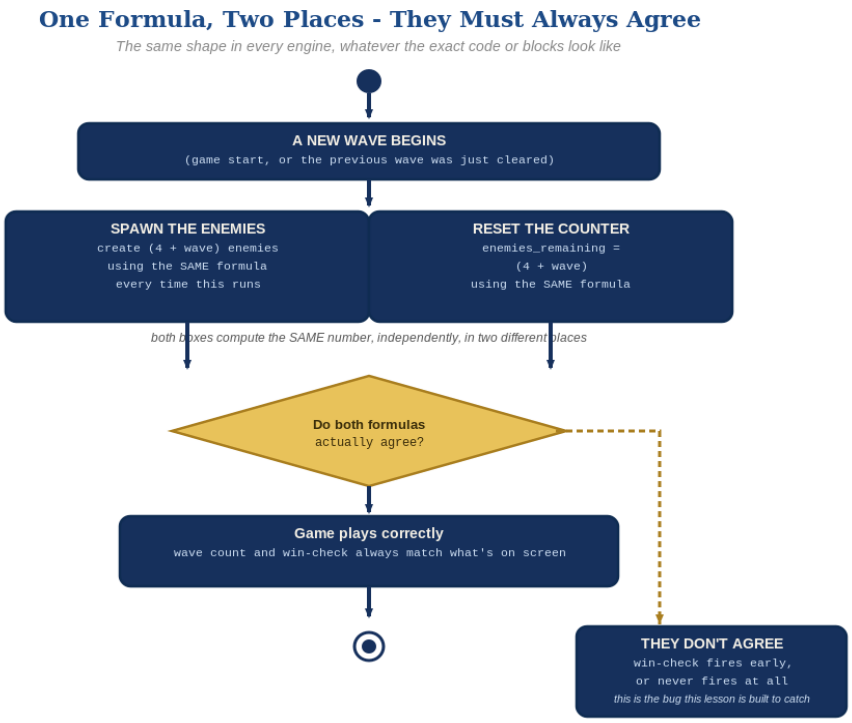
Tip: no new assets this week - difficulty scaling reuses your exact existing enemy sprites and spawn logic.

Setup

1. Open your [YourName]Game project from Lesson 13.
2. In the Map editor, copy-paste 2 more enemy copies from your existing row, naming them enemy_6 and enemy_7 and keeping at least a full enemy-width of gap from their neighbors, same as your original row from Lesson 5 - you now have 7 total pre-placed copies to work with (this engine's enemies are always pre-placed, never runtime-spawned, so growing the count means adding more copies now, at design time).

Level 1 - Minimal (MVP)

Each wave spawns one more enemy than the last (5, 6, 7). Speed stays the same for now - that's Level 2.



Whether it's two places (most engines) or the loop and the reset living in genuinely different scripts (Godot, GameMaker), the rule is the same: one formula, referenced everywhere.

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Whether the spawn loop and the reset live in two places or two genuinely different scripts, the rule is the same: one formula, referenced everywhere.

3. In your ResetGame Custom Event, add Actor – Destroy/Revive(Destroy) for enemy_6 and enemy_7 - by default, only the wave 1 count (5) should be active.
4. In SpawnWave, add an If(wave >= 2) branch that also revives/repositions enemy_6, and an If(wave >= 3) branch that also revives/repositions enemy_7 (code below) - each wave activates one more pre-placed copy than the last.
5. Update enemies_remaining's reset value in both ResetGame and the win-check's Else branch to match: 4 + wave, instead of a fixed 5.
6. Save, then Play/Test and play through a full round - wave 2 should visibly show enemy_6 join in, and wave 3 should show enemy_7 join in too.

Highlighted lines are new this week; everything else is code you already have.

```
Custom Event: SpawnWave (revised) ->
  Actor – Destroy/Revive(Revive) + Teleport(enemy_1..enemy_5, starting
positions)
  If(wave >= 2)
    Actor – Destroy/Revive(Revive) + Teleport(enemy_6, starting position)
  If(wave >= 3)
    Actor – Destroy/Revive(Revive) + Teleport(enemy_7, starting position)

Custom Event: ResetGame (revised) ->
  Variable – Modify Value(wave, Set, 1)
  Variable – Modify Value(enemies_remaining, Set, 4 + wave)
```

```
  Actor – Destroy/Revive(Destroy) for enemy_6, enemy_7
  Call Custom Event(SpawnWave)
```

Level 2 - Complete (Alpha/Beta)

Enemy speed ALSO increases each wave, on top of the growing count. Restarting correctly returns everything to wave 1's original values.

- This engine's confirmed actions don't include a verified way to change Automatic Movement speed at runtime - if your installed version exposes one (check the enemy Template's action list for a Movement-related action), scaling speed per wave is a great Level 3 stretch instead of a required Level 2 item here.
- For this engine, Level 2 instead means: confirm all 7 pre-placed copies revive/reposition correctly across a full 3-wave playthrough, with no copy left stuck mid-Destroy or appearing in the wrong wave.
- Restart mid-game and confirm enemy_6 and enemy_7 correctly return to Destroyed until their wave is reached again.

Level 3 - Refined - Choose ONE Stretch Feature

Pick exactly one of the five ideas below. These are intentionally open-ended - a good-faith attempt at the general idea is a complete Level 3 finish, not a fully polished feature.

- Destructible shields/barriers: a few simple blocks placed between the player and the enemies that disappear (or shrink) after absorbing a hit or two - reuses your existing collision-detection and hit-reaction patterns from Lessons 4 and 13, just on a new kind of object.
- A power-up: an item that occasionally drops or spawns, which the player collects to temporarily change how shooting works (faster fire rate, multiple bullets at once, or a brief shield) - reuses your existing timer/cooldown pattern from Lesson 3.
- A bonus enemy: a rare, faster-moving, or oddly-patterned enemy worth extra points, spawned occasionally alongside a normal wave - reuses this week's own speed/count scaling ideas, just for one special case instead of the whole wave.
- Formation-marching movement: instead of each enemy independently drifting down (this course's approach since Lesson 4), have the whole wave move together as a group and occasionally step sideways - a genuinely different movement pattern, inspired by classic arcade shooters.
- A title screen and/or game-over screen: a proper opening screen (press a key to start) and/or a dedicated game-over screen with a final score, instead of just text overlaid on the gameplay - reuses your existing game_active flag and UI tools from Lessons 9-10. 001 Game Creator's Interface/Field system (already used for your HUD since Lesson 9) is a natural fit for a title/game-over screen.

If Something's Wrong

- enemy_6 or enemy_7 never appears: confirm their SpawnWave If-branch conditions use >= (not just =), so they stay active once revived, not just on the exact wave they're introduced.
- "You Win!" fires early or never: confirm enemies_remaining is reset using the SAME (4 + wave) formula everywhere it's set.

- Restarting leaves enemy_6/7 visible from a previous game: confirm ResetGame explicitly Destroys them before SpawnWave runs.

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